Problem Chosen	2024	Team Control Number
B	**MCM/ICM Summary Sheet**	**7391861**

Finding a Drifting Submersible Before Recovery Becomes Impossible

After a missed contact, MCMS must update both location and detectability. We recommend independent emergency acoustics, depth/fault telemetry and survival-aware planning. Approval also requires recovery equipment compatible with operating depth; search capability alone cannot protect passengers.

We construct a hybrid location ensemble for three fault regimes: a negatively buoyant vehicle that settles on sloping seabed, a suspended vehicle advected by uncertain currents, and a vehicle whose propulsion continues after communication fails. Persistent current bias produces uncertainty proportional to elapsed time, whereas diffusion grows only with its square root. At six hours, a conditional neutral-drift example has eastward standard deviation 557 m; improved current information and a 25 m last fix reduce this to 236 m.

The location ensemble supplies feasible survey cells and depths. A gain-per-time route builder charges ship transit, descent, movement, surveys and a return reserve. Non-detections reduce surviving particle weights; cumulative discovery is one minus their total. In a synthetic 3,000 m deep Ionian scenario, survival-aware discovery is **70.8% at 6 h** and **81.2% at 18 h**, compared with 26.1% and 50.2% for a prior-informed fixed lawnmower under identical conditions. Restricted mean time through 18 h falls from **11.4 h to 6.5 h**. A same-domain cumulative-coverage comparator reaches 70.2% at 18 h and restricted mean 7.7 h, leaving an 11.1-point advantage. This narrower comparison includes probabilistic ordering and other heuristic differences; misses-only routes remain precompilable. A stronger, rotated-current challenge reduces 18 h discovery to 47.3%.

Readiness is evaluated through annual ownership, maintenance, standby and deployment costs together with available life support. Exact procurement prices are unavailable, so recommendations use capability limits and a transparent break-even inequality. The host carries rapid acoustic tracking; a contracted deep search/intervention package supplies compatible launch, navigation, operators and recovery interfaces. Published examples include a 6,000 m survey AUV, a 6,500 m intervention ROV, and a docking rescue system limited to 610 m.

Regional extrapolation replaces environmental forcing and bathymetry rather than retaining Ionian coefficients. Multiple submersibles require separate identities, shared current uncertainty, acoustic slots and joint resource assignment. A small exhaustive allocation check achieves 0.975 expected discoveries with 2.0 fleet-hours by searching different cells.

Alternative forcing and persistent sonar failure preserve a risk floor after repeated misses. Results remain conditional; local acoustic and recovery trials are essential. **Keywords:** hybrid drift; Bayesian search; persistent misses; readiness; deep rescue.

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1 Decision scope and environmental information

The company's safety question has two distinct endpoints: locating the submersible and bringing its occupants to safety. We predict a joint position distribution, use it to dispatch search equipment, and require recovery capability before approval. A location probability alone cannot establish survivability.

The original problem supplies no hull geometry, incident coordinates, measured currents, failure frequencies, sonar trials, costs or life-support duration [1]. Our numerical case is therefore a conditional deep-Ionian scenario, specified in Table 1. A real operation would replace its forcing with an archived forecast and local measurements. Copernicus supplies Mediterranean temperature, salinity and currents on a roughly 4–5 km grid with 141 depth levels [3]; such a product is an environmental prior, while local acoustic/CTD observations must resolve the incident area. Density and sound speed should be derived consistently from temperature, Absolute Salinity and pressure using TEOS-10 [2].

A widening circle is an inadequate baseline for location: it loses directional drift, depth and the stopping effect of seabed contact. Full computational fluid dynamics would demand unsupplied boundary and hull data. We instead use physical equations to define regimes and propagate their uncertainty through a modest ensemble. The scientific claim is conditional policy performance, with physical and computational checks stated separately. AI assisted the formulation, code and entire written report [10]; its actual role is disclosed after the solution.

Table 1. Conditional deep-Ionian scenario

Quantity	Specification
Last fix and seabed	$(0, 0, 2800)$ m; horizontal s.d. 100 m, depth s.d. 15 m; $H = 3000 + 0.025x + 0.010y$ m
Current and diffusion	Mean $(0.050, 0.020)$ m/s; fixed-bias s.d. $(0.025, 0.015)$ m/s; $K = 0.2$ m^2/s
Fault prior	Sinking/neutral/communication-only: $(0.50, 0.40, 0.10)$; sinking speed $U[0.03, 0.12]$ m/s
Suspended and propelled	Neutral-depth s.d. 35 m, relaxation 900 s; residual propulsion 0.06 m/s, heading s.d. $\pi/3$ rad
Travel and preparation	Host 5.0 m/s; sensor horizontal 1.8 m/s, vertical 0.5 m/s; preparation 0.5 h
Endurance and return	20 h wet duty-cycle endurance; 20% reserve; descent, search and return must fit 16 h
Survey and detection	Cell side 500 m; survey 0.25 h; depth tolerance 100 m; inside-cell hazard $8q$ h^{-1}, $q \in \{0.4, 0.7, 0.9\}$ equally weighted
Numerics	2,400 trajectories; 120 s drift step; five midpoint detector samples per survey; seed 7391861; horizon 18 h

2 A hybrid three-dimensional location model

2.1 Buoyancy, fault states and seabed contact

Use east and north coordinates $\mathbf{x} = (x, y)$ in metres and depth $z \geq 0$ positive downward. The local water depth is $H(\mathbf{x})$. Before contact, a minimal vertical force balance is

$$m_{\text{eff}}\dot{w} = g\{m - \rho(S_A, \Theta, p)V(p)\} - \tfrac{1}{2}\rho C_D A_z w|w|, \qquad \dot{z} = w. \tag{1}$$

Here m_{eff} includes added mass, m is vehicle mass, V displaced volume, A_z the vertical projected area and C_D a drag coefficient. Both forces have units N. Leakage, ballast state and pressure-dependent hull volume alter the first term. If they are nearly constant, the downward terminal speed is

$$w_\infty = \operatorname{sgn}(m - \rho V)\sqrt{\frac{2g|m - \rho V|}{\rho C_D A_z}}. \tag{2}$$

Without hull parameters, selecting a sinking-speed range is more honest than inventing them. A neutral equilibrium requires $m = \rho V$, but stable equilibrium additionally requires $\partial_z[g(m-\rho V)] < 0$. Water compression and hull compression compete; density equality by itself does not guarantee the vehicle remains at a fixed depth.

We condition on a discrete fault state R: sinking with lost propulsion; suspended with lost propulsion; or communication failure with residual commanded propulsion. The sinking ensemble uses Equation (2)'s terminal approximation. The suspended states use $\dot{z} = -(z - z_*)/(900\,\text{s})$, representing stable trim near an uncertain equilibrium depth z_*. An unstable or positively buoyant state belongs in a separately assessed ascending branch. Reaching the surface changes the sensor to radio/GNSS/visual search; it is not represented as a deep neutral state. Thus the numerical case is conditioned on nonascending incidents, while the operational model supports all buoyancy signs.

At $z = H(\mathbf{x})$, an inelastic contact transition sets $w = 0$. Our principal scenario assumes the vehicle sticks. This is appropriate only if bottom friction and terrain resist both current drag and downhill loading. Let the submerged normal load be N and bottom slope angle α. A sufficient sticking condition is

$$\|\mathbf{F}_{\text{drag},\parallel} + \mathbf{F}_{\text{gravity},\parallel}\| \leq \mu N, \qquad N > 0. \tag{3}$$

If buoyancy nearly cancels weight, N is small and sticking can fail. Then a bottom-slip branch projects velocity onto the seabed tangent plane, with friction and obstacle constraints; an inspection sonar and a repeated bottom position are needed to distinguish it. No particle may pass through seabed or a coastline. At the surface, depth is clamped at zero and the state changes to a surface drift/search model. The scenario's inclined plane contains no land or cliff; operational bathymetry must add those boundaries.

2.2 Current uncertainty and the location distribution

For a mobile vehicle, write

$$d\mathbf{x} = [\overline{\mathbf{u}}(\mathbf{x}, z, t) + \mathbf{b} + \mathbf{v}_R]\,dt + \sqrt{2\mathbf{K}}\,d\mathbf{W}_t. \tag{4}$$

The archived forecast supplies $\bar{\mathbf{u}}$, while a trajectory-specific bias $\mathbf{b}$ represents persistent forecast error. $\mathbf{v}_R = 0$ for propulsion loss; a heading distribution represents continued propulsion when communication alone fails. Diffusion summarizes unresolved fluctuations, not uncertainty about the same fixed current parameter. In the constant-current neutral limit,

$$E[\mathbf{x}(t)] = \mathbf{x}_0 + \bar{\mathbf{u}}t, \quad \Sigma_x(t) = \Sigma_0 + t^2\Sigma_b + 2\mathbf{K}t. \tag{5}$$

All terms in Σ_x have units m^2. Redrawing $\mathbf{b}$ at every small step incorrectly converts persistent error into a random walk and understates long-term dispersion.

Particles sample the joint last fix, fault state, equilibrium/sinking speed, current bias, heading and detectability. Propagation gives

$$\pi_t(d\mathbf{x}, dz, dR, dq) = \sum_{i=1}^{N} p_i \, \delta_{(\mathbf{x}_i(t), z_i(t), R_i, q_i)}. \tag{6}$$

Fault probabilities are scenario weights, not estimated accident frequencies. Marginal uncertainty combines distinct depth modes; a single covariance ellipse can summarize horizontal containment but cannot select a survey depth. Figure 1 shows the divergence: bottom targets stop, whereas suspended and still-propelled targets continue to spread. The empirical 95% horizontal ellipse grows from 0.95 km^2 at 2 h to 8.2 km^2 at 6 h and 73.4 km^2 at 18 h. These ellipses are ensemble containment summaries, not calibrated confidence regions.

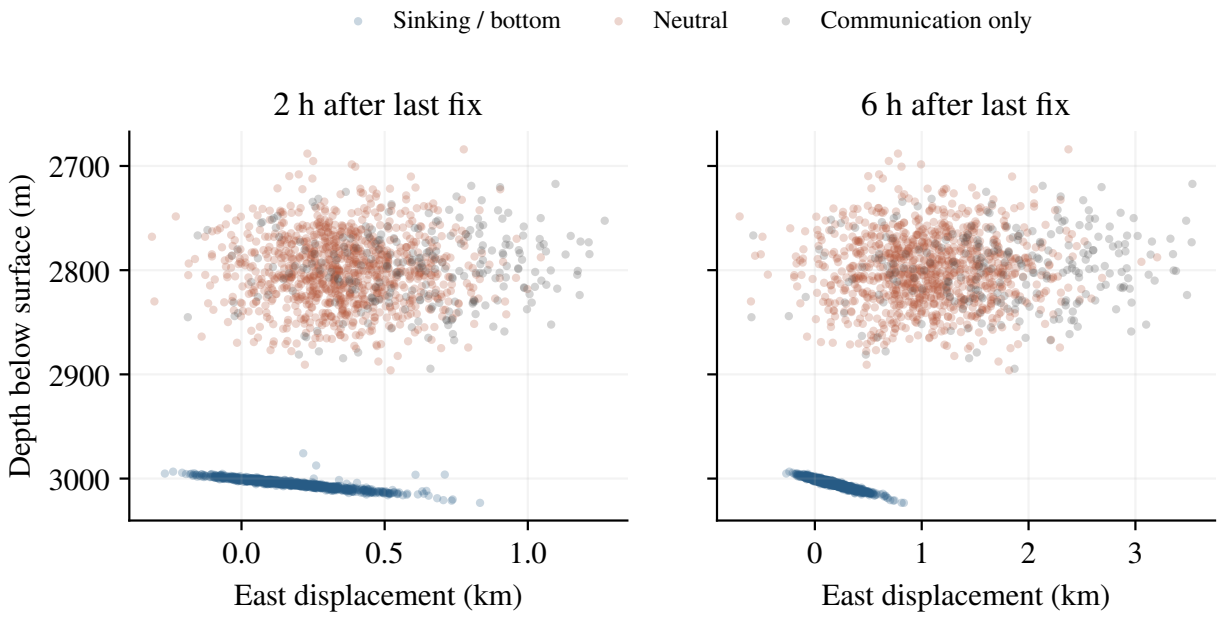


Figure 1. Fault-dependent depth and drift

3 Telemetry that changes the search decision

3.1 Observable states and useful equipment

Each packet should carry synchronized time, vehicle identity, pressure depth and depth rate, position estimate with covariance, heading/attitude, ground- or water-referenced velocity, thruster commands and fault flags, ballast/leak alarms, remaining battery, oxygen and carbon-dioxide margins. A pressure sensor/altimeter distinguishes suspension from bottom contact; DVL and inertial navigation track motion between acoustic fixes; CTD/sound-speed profiling improves buoyancy and acoustic ranging. DVL loses bottom lock outside its working altitude, so an unaided inertial fix needs a growing covariance rather than a precise-looking coordinate.

The host combines GNSS and heading/attitude data with calibrated ultra-short baseline (USBL) ranging. Ranger 2 advertises up to 11 km range and optimized system accuracy of 0.04% of slant range, with two-way telemetry [4]. These maxima require geometry, sound-speed correction, calibration and suitable acoustics. They do not justify centimetre or metre accuracy in an untested tourist operation. A second emergency pinger needs its own battery and protected activation path, since a beacon sharing propulsion power can fail in the same incident.

A reported position without time, covariance or sensor status is insufficient. A USBL fix and an inertial/DVL estimate may share calibration errors; fusing them as independent measurements overstates information. Conditional on trusted measurement y, update particles with $p_i^+ \propto p_i^- L(y \mid X_i, R_i)$. Sensor faults belong in L, and a missing packet is first a communication observation, not proof of sinking.

3.2 Cadence and the value of current information

At incident declaration, use the actual last-valid-fix age $A = t_{\text{declare}} - t_{\text{last fix}}$, including reception and processing delay. With scheduled reporting interval Δ and no intervening valid packet, one missed report gives $A = \Delta + \ell$ and two missed reports give $A = 2\Delta + \ell$, where $\ell \geq 0$ is the declaration delay. Derive $r_{95}(A)$ from Equation (5); accept the reporting/trigger rule only if $r_{95}(A) \leq r_{\text{acquire}}$ under conservative forecast errors. A delayed received fix is aged from its measurement timestamp. Packet and trigger delays must also remain within safe depth/ballast response time. No universal interval follows without local trials.

Equation (5) gives a useful purchasing insight. At six hours, the scenario's neutral eastward standard deviation is $\sqrt{310240} = 557$ m. Reducing the last-fix error alone does little when current error contributes 540^2 m^2. Reducing current-bias s.d. to 0.010 m/s and last-fix s.d. to 25 m gives 236 m. Better current information is therefore more valuable here than polishing an already reasonable coordinate.

Fault telemetry can be equally decisive. For an illustrative sinking-status bit with likelihoods $(0.90, 0.10, 0.15)$ under the three regimes, a positive report has probability 0.505 and makes the sinking posterior 0.891; a negative report makes it 0.101. Averaging both possible reports reduces regime entropy from 1.361 to 0.844 bits. This is a before-observation information calculation, not

evidence that an actual sensor has those error rates or that its search-time benefit equals its entropy reduction. Local trials should estimate its likelihood and compare the resulting dispatch policies.

4 Preparation, readiness and recovery

4.1 Layered capability on the host ship

The immediate host package is calibrated acoustic tracking/telemetry with redundant emergency reception, spare beacon batteries and transducer, CTD/sound-speed profiling, navigational logging, deployment-safe lifting and trained watchkeepers. Its job is to preserve or reacquire contact before uncertainty spreads. A small inspection ROV may serve shallow operations, but its pressure rating must cover the authorized depth.

For deep operations, the second layer is a depth-rated search vehicle with bottom imaging and independently tested midwater detection, terrain avoidance, position aiding, sufficient endurance, launch/recovery hardware and rapid access to observations. HUGIN 6000 provides an example of a 6,000 m platform: the published range is 2–5 knots, with 240 m synthetic-aperture sonar range at 3 knots and 2–2.5 km^2/h advertised coverage [5]. A seabed imaging swath is not a guaranteed midwater detection volume. Our 500 m cell detector is an explicit effective-survey hypothesis that must be calibrated with a known target at different altitudes, attitudes, slopes and sound-speed profiles.

An autonomous vehicle can return too late to guide an urgent search. To implement our 15-minute replan, a vehicle must send target/miss summaries promptly or be supervised through a suitable acoustic link; raw high-resolution imagery may require retrieval. Observation latency is therefore a capability requirement. Without it, execute a longer preplanned sortie and propagate the posterior only when actual data arrive.

4.2 Lifecycle costs and a readiness threshold

For package e, an annual equivalent cost is

$$C_e = \mathrm{CRF}(i,n)P_e + M_e + T_e + R_e + \nu(C_{\mathrm{mobilize,e}} + c_{\mathrm{ship,e}}h_{\mathrm{ship}} + c_{\mathrm{use,e}}h_{\mathrm{use}}), \tag{7}$$

where P_e is purchase price, M_e maintenance/consumables, T_e training/calibration, R_e standby/retainer and ν annual incident/activation frequency. $\mathrm{CRF} = i(1+i)^n/((1+i)^n - 1)$ converts capital to annual cost. Quotes should specify depth options, batteries, service intervals, operators, vessel compatibility, export/transport constraints and emergency response commitment. No verified MCMS purchase or charter prices were available. A quote range $[C_e^-, C_e^+]$ and tested availability range $[a_e^-, a_e^+]$ give an interval decision. A historical primary example shows the scale of readiness obligations: the UK Ministry of Defence reported a three-year NSRS support extension worth more than £7 million in 2020, covering trained teams, upgrades, full maintenance and year-round readiness [8]. This is a national support contract, not a current MCMS equipment price or fee per rescue.

Let $D = L - t_{\text{extraction}} - t_{\text{reserve}}$ be the latest useful discovery time from life support L. Availability a_e includes serviceability and access; delay d_e includes mobilization and voyage. The useful discovery chance is $a_e F_e(D - d_e)$, zero for a nonpositive argument. An additional package is economically preferred to a reference if

$$\Delta C \leq vV\{a_e F_e(D - d_e) - a_0 F_0(D - d_0)\}, \tag{8}$$

where V is the decision maker's consequence valuation, kept explicit. Here F_e must be recomputed for deployment at delay d_e: a moving target's incident clock is not reset by mobilization. The compact notation describes useful search time; it does not license shifting our existing curve without propagating the target during the delay. Safety requirements may impose a minimum readiness independently of that valuation. A separate normalized comparison, 0.95 availability with a twelve-hour decision horizon versus 0.70 with six hours, gives increment 0.264 using the same scenario curve. Its annual break-even increment is $0.264vV$, before extraction compatibility and common-cause failures. A more capable tool that arrives after D has no rescue value in this calculation.

Onboard deep AUV ownership is justified only when emergency access/launch delay cannot be met by a binding contract and deck/crew requirements can be maintained. Otherwise the proportionate plan is to own the rapid acoustic package and contract a verified deep search/intervention vessel. "Available nearby" is not a readiness metric: agreements need mobilization times, vessel-of-opportunity interface checks, maintenance status and rehearsals.

4.3 What the rescue vessel must add

A rescue vessel must bring depth-rated intervention imaging, manipulators, compatible tooling, strong station keeping, launch winch/crane and cable, navigational/acoustic support, trained crews, medical care and a verified recovery or transfer interface. WHOI's Jason is an example of a 6,500 m intervention ROV with cameras, sonar and manipulators [6]; its scientific manipulation capability does not establish that it can lift a human submersible. Recovery needs certified lift points and rigging, load and buoyancy estimates, seabed clearance, controlled ascent and surface lifting limits.

The NATO Submarine Rescue System's published docking rescue depth is 610 m; its vehicle carries up to 15 passengers and supports transfer under pressure [7]. Its pressure rating, hatch geometry and mobilization are separate constraints. It cannot be presumed to recover occupants from our 3,000 m scenario, nor assumed to be commercially available to MCMS. At such a depth, a company would need a tested emergency ascent or a specifically engineered lifting/recovery route with intact hull and viable occupants. If neither is credible within life support, the regulator should restrict the operating depth or refuse that operation even if the search model is excellent.

We also integrate search with a conditional recovery clock. All times below are hours, depth z is metres and ascent speed v_{ascent} is m/s. If contact occurs at t, intervention takes I_R and the resource arrives at d_r, the earliest modeled surface completion is

$$c(t, X, R) = \max(t, d_r) + I_R + \frac{z}{3600 v_{\text{ascent}}} + h_{\text{surface}}. \tag{9}$$

Depth, certified interface and hull condition must all permit the task. Using the survival weights defined in Section 5, the timing/depth opportunity for package e is

$$Q_e = \sum_i \int_0^H \lambda_i(t) s_i(t)\, \mathbf{1}\{c(t, X_i, R_i) \le L - r,\ z_i \le z_{\max,e}\}\, dt. \tag{10}$$

Actual recovery multiplies the integrand by conditional availability and intervention success $r_e(X_i, t)$, which are unmeasured here. Consequently Q_e is an upper opportunity bound under the stated clock, not the probability of saving passengers.

For an illustrative 18 h life support, 2 h reserve, rescue arrival at 6 h, intervention times 3/5/4 h for sinking/neutral/communication-only states, ascent 0.25 m/s and 1 h surface handling, an already-compatible 6,000 m recovery package yields continuous timing/depth opportunities of 75.4%, 67.3% and 41.4% for survival-aware, cumulative-coverage and compact-lawnmower search. The first and last are the original routes; the middle is the matched-domain comparator developed in Section 6. The same calculation gives zero eligibility for a 610 m docking tool. These hypotheses demonstrate the benefit a genuinely compatible rescue ship could add, while exposing the failure of depth-incompatible equipment. They do not establish an engineered deep extraction method or its success rate.

5 Search driven by the surviving location distribution

5.1 Detection and the meaning of a negative survey

Let a survey action a cover a 500 m square cell and one depth band. Define $K_a(X_i(t)) = 1$ inside the footprint and within 100 m of survey depth, otherwise zero. Its conditional hazard is $\lambda_i(t) = 8 q_i K_a(X_i(t))$ h^{-1}. The latent q_i is fixed throughout an incident: poor reflectivity, terrain shadow or an unsuitable sensor may cause correlated misses. It is not redrawn after each pass.

Let $E_{ia} = \int_{t_a}^{t_a+\tau_a} K_a(X_i(s))ds$ in hours. Beginning with weights $s_i(0) = p_i$, non-detection multiplies them by

$$s_i^+ = s_i^- \exp(-8 q_i E_{ia}), \qquad S(t) = \sum_i s_i(t), \qquad p_i(t \mid T > t) = \frac{s_i(t)}{S(t)}. \tag{11}$$

The posterior retains joint location, fault and detectability uncertainty. Motion continues while the equipment travels; only the survey interval has detection hazard. This is consistent with the cumulative success and particle-failure perspective in operational search planning [9].

An exact check illustrates the effect: if cells A/B initially have masses 0.6/0.4 and one survey detects A with probability 0.8, a miss leaves total survival 0.52, with A posterior $0.12/0.52 = 3/13$. Repeating A then offers conditional success $(3/13)0.8$, rather than the original 0.48. A successful acoustic or imaging candidate receives a likelihood update and an identification/inspection task. False alarms and identity uncertainty require confirmation; the reported first-discovery curve assumes the effective hazard has already been calibrated for correct identification.

5.2 Deployment and feasible survey routes

At time t, a candidate $a = (x_a, y_a, z_a)$ has movement time

$$d_a = \frac{\|\mathbf{x}_a - \mathbf{x}_{\text{sensor}}\|}{v_h} + \frac{|z_a - z_{\text{sensor}}|}{v_z}, \qquad t_a = t + d_a. \tag{12}$$

Speeds here are expressed in m/h. This conservative orthogonal movement rule is common to both policies. The initial move uses host speed 5.0 m/s before descent; later moves use sensor horizontal speed 1.8 m/s. Sensor vertical speed is 0.5 m/s. Preparation is charged before deployment. A 0.25 h effective survey includes local coverage and turns; there are no uncharged observations during transit.

The finite-horizon objective is $\min_\pi E[\min(T, H)]$, $H = 18$ h, subject to travel, survey depth, observation timing and an implemented duty-cycle endurance/return constraint. We assume 20 h wet endurance and protect a 20% reserve. Each candidate must satisfy $t_a+\tau_a+d_{\text{return}}(a) \le t_{\text{launch}}+16$ h, using the same horizontal/vertical movement rule to return to the host's actual launch location. This proxy requires endurance trials at the proposed survey/transit duty cycle; it is not a physical power-consumption model. Our feasible approximation scores a cell at its future arrival time:

$$G_a = \sum_i s_i(t)\{1 - e^{-8q_i E_{ia}}\}, \qquad a^* = \arg\max_{a:\, t_a < H} \frac{G_a}{d_a + \tau_a}. \tag{13}$$

It charges travel and allows revisits when residual mass warrants them. After each completed negative survey, Equation (11) updates the distribution and the next cell is selected. An 80-candidate shortlist uses the posterior at $t + 0.5$ h, weighted by q, from a 500 m lattice spanning $x \in [-2, 8]$ km and $y \in [-4, 5]$ km. Each candidate then receives a future-arrival calculation; cells outside the lattice retain probability but are not credited as searched. A field implementation should expand the lattice if residual mass accumulates outside it.

This heuristic is locally motivated, not a global optimum of a moving-target routing problem. Its value must be demonstrated against a feasible baseline. The first deployment is $(0, 0)$ horizontally, with survey depth 2,960 m. Preparation and descent delay the first survey until 2.144 h. The next survey is at $(500, 0, 2800)$ m, beginning 2.560 h. These are locations relative to the last fix, not unverified geographic coordinates.

Proposition 1. *For disjoint single-visit cells with no travel or drift, mass p_j, detection probability d_j and duration τ_j, with discovery reported at completion, set $H = \sum_j \tau_j$. An order with nonincreasing $p_j d_j/\tau_j$ minimizes $E[\min(T, H)]$.*

Proof. Let discovery be reported at a cell's completion time C_j. The restricted expectation is $\sum_j p_j d_j C_j + (1 - \sum_j p_j d_j)H$. For adjacent cells i, j, the order i, j instead of j, i changes this objective by $p_j d_j \tau_i - p_i d_i \tau_j$, nonpositive exactly when $p_i d_i/\tau_i \ge p_j d_j/\tau_j$. Adjacent swaps give the order; the undiscovered mass contributes the same fixed H. This avoids an unconditional expectation that may be infinite when discovery fails. Travel, drift and repeated surveys break separability, hence Equation (13) is an approximation in our full problem. □

5.3 Probability over elapsed time and accumulated results

On the all-negative branch of a policy, the cumulative probability of first discovery is

$$F_\pi(t) = P(T \le t) = 1 - S_\pi(t), \qquad E[\min(T, H)] = \int_0^H S_\pi(t)\, dt. \tag{14}$$

This branch is sufficient: before the first discovery, the history contains only misses, so it uniquely determines each next action. A success terminates first-discovery timing. Consequently the entire all-negative route can be precompiled at launch with the same cumulative distribution. Our comparison measures the value of survival-aware route construction relative to a specified lawnmower, not an independent advantage of online computation. New environmental measurements, nonterminal candidate contacts, changed sensor status or fault telemetry create genuinely different information branches and justify online replanning. Conditional on having searched until t without success, the chance of discovery before $t + u$ under the continuation is

$$P(T \le t + u \mid T > t) = 1 - \frac{S_\pi(t + u)}{S_\pi(t)}. \tag{15}$$

Thus a cumulative 80% prior probability is not an 80% chance of imminent success for the crew after repeated misses. In our scenario, after no discovery through six hours, the conditional chance of finding the vehicle by twelve hours is 31.3%. Equation (15) explicitly uses accumulated results. If a contact, new current profile or fault report changes the information, reweight/propagate the joint distribution and replan from the actual equipment location; past elapsed time and used endurance do not reset.

6 Conditional performance and independent checks

6.1 Comparison under common physical and information conditions

The fixed reference begins in the same modal bottom cell as the survival-aware policy. It then follows complete lawnmower rows over $x = -500, 0, 500, 1000, 1500$ m and $y = -500, 0, 500, 1000$ m, alternating a bottom and a suspended-depth row. The field approximates the compact deployment-time envelope; it repeats if time allows. Both policies receive the same prior ensemble and detector, preparation, ship/sensor speeds, survey effort, endurance/return reserve and observation assumptions. The fixed reference processes misses in its probability calculation but retains its preplanned route. The comparison changes route construction, including drift coverage and survival-weighted ordering.

Table 2 and Figure 2 show the resulting first-discovery probabilities. Survival-aware search front-loads high surviving mass and follows drifting suspension, reducing restricted mean time by 4.9 h. Its 18 h probability gain is 31.0 percentage points against this compact reference; the same-domain comparison below narrows that gap. These values do not establish dominance over all fixed patterns or global optimality. Search ends at 14.4 h and 14.3 h; the two vehicles return by

16.36 h and 16.30 h, with 20% endurance reserve protected. Their curves are flat during return and afterward. A differently calibrated prior or endurance can require another route.

Table 2. Search performance

Policy / environment	$F(6\,\mathrm{h})$	$F(12\,\mathrm{h})$	$F(18\,\mathrm{h})$	Restricted mean (h)
Survival-aware / nominal	70.8%	79.9%	81.2%	6.5
Lawnmower / nominal	26.1%	50.1%	50.2%	11.4
Survival-aware / challenge	43.3%	47.0%	47.3%	11.2
Lawnmower / challenge	23.9%	46.0%	46.0%	12.0

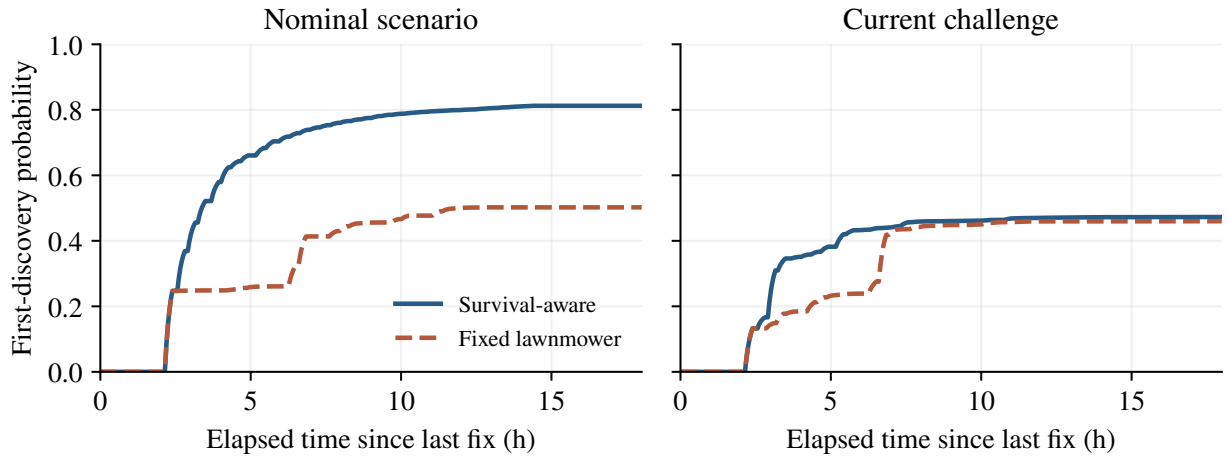


Figure 2. Cumulative discovery by environment

Figure 3 illustrates why a systematic pattern can spend valuable time on low-probability cells. The survival-aware route changes depth early after bottom misses; all moves consume time. Squares denote bottom surveys and circles suspended surveys, over the 2 h horizontal ensemble density; selected numbers give order. At 18 h without discovery, posterior regime masses are 5.0% sinking, 52.1% neutral and 43.0% communication-only. The rare moving fault state becomes important because the easy bottom mass has mostly been searched.

The fixed field contains 100% of horizontal prior mass at 2 h but only 52.8% at 18 h. The larger candidate lattice contains 98.0% at 18 h; the two depth bands contain 99.8%, and their joint support contains 97.9%. On the survival-aware all-negative posterior, 10.4% lies outside the lattice and 0.9% outside the depth bands; the union is 11.3%. This mass remains in $S(t)$ rather than being silently discarded. Thus the gain over the particular lawnmower includes following the drifting tail.

A prior-only dynamic ablation uses the same lattice, future-arrival scoring, movement, detection and endurance, but never downweights previously missed trajectories for action selection. Its actual discovery calculation still accumulates misses. It repeatedly favors the modal bottom cell, reaches 34.6% at 18 h and has restricted mean 12.6 h. Survival-aware selection avoids this repeated allocation. This supports using surviving mass in route construction; it does not measure independent online information value, because each all-negative route remains precompilable.

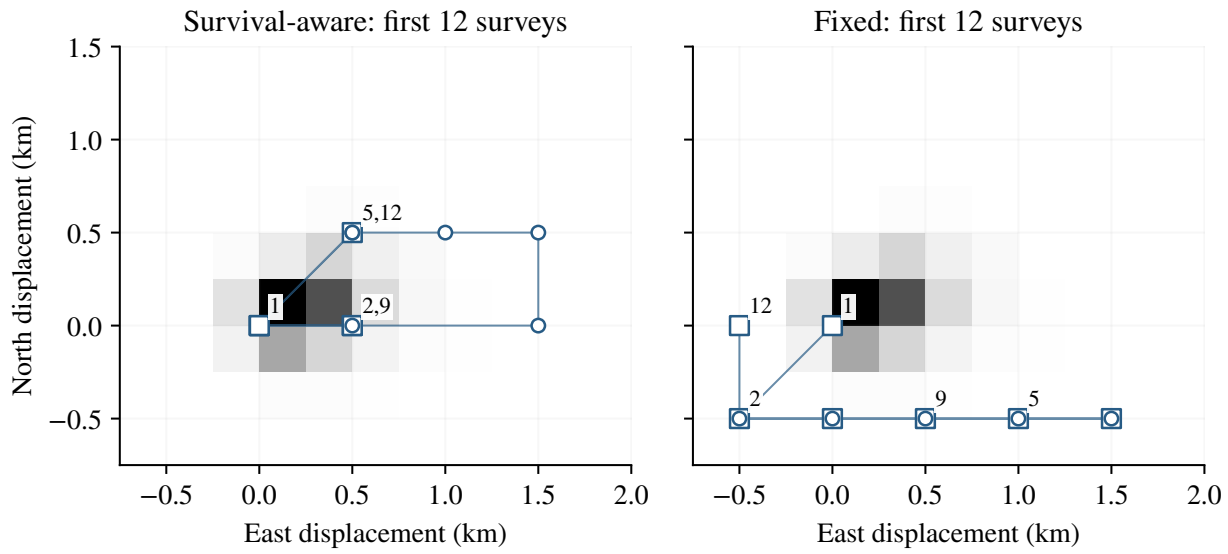


Figure 3. Initial survey routes

6.1.1 A time-aware cumulative-coverage comparator

The prior-only ablation can revisit the modal cell indefinitely, so it is a weak substitute for systematic coverage. We add a comparator that remembers survey geometry but does not use negative-observation probabilities to choose its next action. It shares the survival-aware policy's 798 candidate cell/depth actions, 80-action shortlist mechanism, future-arrival evaluation, last-fix ensemble, initial bottom deployment, preparation, speeds, detector, 18 h horizon and 16 h usable wet endurance including actual return. The compact lawnmower is retained as a declared route within this physical domain; its smaller coverage remains part of that policy's definition.

Let C_i be trajectory i's cumulative geometric survey exposure in hours and $\tau = 0.25$ h the already specified survey duration. For a future action with exposure E_{ia}, score

$$G_a^{\mathrm{cov}} = \sum_i p_i \frac{\min\{E_{ia}, (\tau - C_i)_+\}}{\tau}, \qquad a_{\mathrm{cov}} = \arg\max_a \frac{G_a^{\mathrm{cov}}}{d_a + \tau_a}. \tag{16}$$

The shortlist uses remaining geometric coverage, with no detectability weighting. After a survey, set $C_i \leftarrow C_i + E_{ia}$. A fully exposed trajectory receives no further coverage credit, although a miss need not mean it was detected; this is the comparator's explicit limitation. The one-survey cap comes from the declared effort unit and was fixed before execution, without parameter tuning. Selecting a route by this score is distinct from exponential surviving-mass weighting. All three routes are evaluated afterward with the same persistent q and Equation (11); no false "certain detection after coverage" enters their discovery probabilities.

Table 3 uses independent continuous entry/exit and recovery-clock integration on the same stored linear trajectories; earlier rounded figures use the original detector quadrature. Cumulative coverage substantially strengthens the reference: the survival-aware advantage falls from 31.02 to 11.08 percentage points in final discovery, and from 4.955 to 1.270 h in restricted mean. Its deep timing/depth opportunity advantage falls from 34.00 to 8.02 points. The coverage route makes 30

Table 3. Common-clock comparison on the frozen nominal ensemble

Route	$F(6)$	$F(18)$	Restricted mean (h)	Q_{deep}	Wet + return (h)
Survival-aware	70.87%	81.26%	6.456	75.36%	15.858
Cumulative coverage	65.71%	70.19%	7.726	67.34%	15.827
Compact lawnmower	26.09%	50.24%	11.411	41.36%	15.799

surveys and returns at 16.327 h, leaving 0.173 h inside usable wet endurance. Figure 4 connects discovery with the separate recovery endpoint.

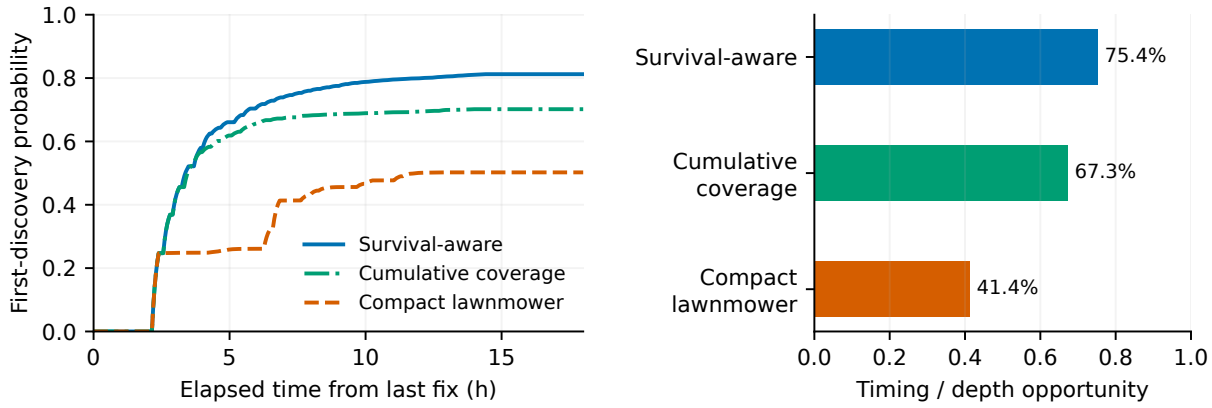


Figure 4. A stronger coverage reference and the recovery clock

This comparison improves the evidence for the recommendation rather than replacing the accepted search route: survival-aware search retains the smaller restricted mean under the declared scenario. The gap against geometric coverage includes graded repeated-detection value, detectability weighting, shortlist and route interactions. It is not an additive causal decomposition of the old 31-point gain, nor a proof of dominance over all equal-domain coverage planners. Both new and old all-negative routes can be precompiled. The forcing challenge and real calibration limits continue to apply.

6.2 Computation checks and their scope

The 25 recorded checks include an analytic constant-current covariance limit, exact rational negative-search arithmetic, depth/contact invariants, independent reconstruction of travel and return endurance, and an independently written whole-pass hazard replay. The latter agrees with the incremental survival computation to floating-point precision. A stronger check solves every entry/exit intersection of each stored linear trajectory with the survey box and integrates exponential survival analytically. It gives $F(18) = 0.81264/0.50239$ and restricted means 6.456/11.411 h for survival-aware/fixed routes; the $F(18)$ discrepancies are below 0.039 percentage points. These checks verify calculation under the shared detector and piecewise-linear trajectory assumptions. The principal absorbed trajectories remain fixed after contact and never penetrate

the analytic seabed.

Restricted mean time is bounded by left/right monotonic rectangle sums around the plotted survival integral; their width is below 0.06 h. The exact event replay independently avoids detector quadrature error on the same trajectories. The added coverage comparison is checked by a separate implementation of movement, all return margins, linear box entry/exit, first-event survival integrals and the continuous recovery deadline. A stationary single-target case has a separately solved exponential integral, and exact rational negative-survey arithmetic checks the conditioning. It reuses the frozen ensemble without regenerating trajectories or rerunning unrelated sensitivity studies. The 120 s propagation discretization and conditional fault/current distributions still limit physical predictions. Real hull, current, acoustic and recovery trials are required for calibration.

6.3 Forcing mismatch and the escalation rule

The challenge doubles eastward mean current to 0.100 m/s, changes northward mean to -0.025 m/s and increases current-bias s.d. to $(0.050, 0.035)$ m/s. It uses independent trajectories and replays the nominal routes without tuning them to hidden target truth. Survival-aware $F(18)$ falls from 81.2% to 47.3%, close to the reference's 46.0%. This is a meaningful warning: a planner can be arithmetically coherent and still confidently search the wrong field.

After repeated misses, obtain a local current/CTD profile, reassess whether the vehicle is slipping, still propelled, above/below the search band or acoustically shadowed, and add an independent modality. Do not merely continue to shrink the same prior. The scenario's detectability mean falls from approximately 0.667 to 0.617 on the all-negative survival-aware branch, showing that failure changes belief about equipment as well as location.

6.4 Persistent non-detectability and model-risk mixtures

Positive detectability alone may exclude correlated failure. We therefore test an explicit 8% prior state in which the selected sonar cannot identify the target, plus a 15% alternative-forcing component. Both are scenario weights for model risk, not calibrated event frequencies. On this mixture, reoptimized survival-aware discovery is **60.5% at 6 h and 69.5% at 18 h**; the fixed reference reaches 46.6% at 18 h. Executing the nominal route on the same risk mixture gives 69.5% at 18 h. Restricted means are 8.155 and 8.112 h, respectively: reoptimization does not improve the mixture objective. On the pure current challenge it improves discovery from 47.3% to 51.5%, a conditional hedge with a nominal tradeoff. We retain the nominal route as the primary comparison and the failed mixture improvement as evidence about this heuristic's limits.

For any persistent $q = 0$ mass η, Equation (11) implies $S(t) \geq \eta$ and $F(t) \leq 1 - \eta$ for this modality, regardless of repeated effort. In the generated ensemble, the blind-state mass is 8.6%. It cannot be removed by another miss. A second sensor's usefulness depends on its joint failure pattern, not only its nominal detection rate. The marginal gain of repeating exposure τ to a stationary uncertain-quality target is $pE[\lambda(q)e^{-\lambda(q)\tau}]$, which decreases with effort. This provides a concrete switch criterion: prefer another feasible cell or independent modality when its gain per

time exceeds this repeat rate.

Replacing persistent quality by its per-pass mean is optimistic. With a 10% invisible state and 90% conditional single-pass detection 0.8, two passes leave survival $0.1 + 0.9(0.2)^2 = 0.136$. Treating each pass as independently detecting with probability 0.72 gives only $(0.28)^2 = 0.0784$. The exact 5.76-point false-confidence gap is why detectability belongs in the posterior. Figure 5 shows the effect on both the discovery curve and the detector belief.

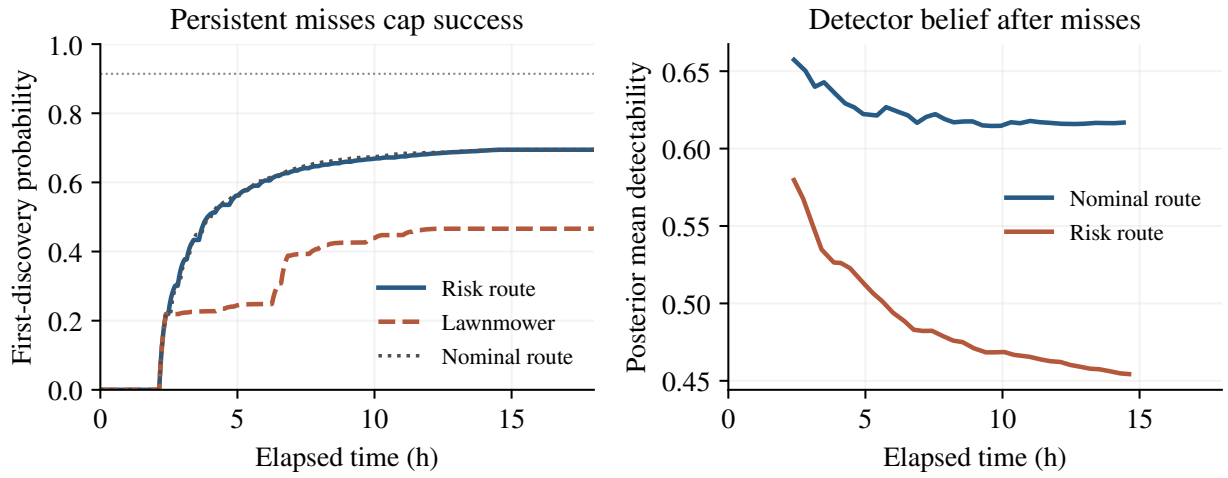


Figure 5. Persistent detection uncertainty

The improved-current/last-fix scenario produces 89.6% survival-aware discovery at 18 h. It is a conditional precision comparison, not the expected value of unknown future telemetry. Repeating the route construction with 1,200 particles and another predeclared seed gives survival-aware/fixed 81.8%/53.0% at 18 h; the nominal 2,400-particle result is 81.2%/50.2%. Replaying the unchanged accepted routes on 9,600 independently drawn same-model particles gives 80.3%/50.8%. The ranking persists, with approximately one percentage point of ensemble variation. Printed decimals are display precision, not calibrated accuracy. Replaying the same routes with 25 detector midpoints changes final discovery by 0.03 and 0.02 percentage points.

Changing the sinking/neutral prior from 0.50/0.40 to 0.20/0.70 or 0.70/0.20, with communication-only fixed at 0.10, yields survival-aware/fixed 82.0%/30.7% and 84.7%/63.9%. The first deployment changes to the suspended layer when its prior dominates. In a bottom-slip challenge with 20% current transfer and 0.005 m/s downslope motion after contact, the nominal route reaches 79.8%; replanning under that known model reaches 81.8%. These are controlled sensitivity comparisons, not access to hidden incident truth. Together they identify forcing, fault weights and bottom behavior as calibration priorities; they do not establish continuous-route optimality or real-world accuracy.

7 Extrapolation to new regions and multiple vehicles

7.1 Caribbean operations as a changed forcing problem

The stochastic and search equations transfer, while their inputs do not. Caribbean deployment requires local current forecasts and depth-resolved observations, bathymetric coastal/reef barriers, density and sound-speed profiles, tide and wind forcing for an ascending/surface branch, and sea-state-dependent launch limits. Land crossings are forbidden; narrow channels and reefs require obstacle-respecting paths. Higher currents can make a fixed search field obsolete before a sortie returns. Regional authority, ports, aircraft/vessel access and recovery agreements determine mobilization rather than the Greek contract.

Our stronger/rotated-current calculation is a conditional Caribbean-type stress test, not a measured Caribbean prediction. The fall to 47.3% at 18 h shows a consequence of the extension. It also justifies retaining several current scenarios and local environmental observations rather than reusing Ionian coefficients. The operating envelope must be requalified for the actual site.

7.2 Identity, interference and joint dispatch

For submersibles $k = 1, \ldots, m$, retain labeled distributions $\pi_t^{(k)}$ and identifiers in every packet/pinger reply. Shared current error couples their positions: safe vehicles can inform the common forcing, but their errors should not be treated as independent. A detection of uncertain identity needs a data-association likelihood across targets; one anonymous contact cannot count as finding all vehicles.

Each sonar/USBL exchange consumes an acoustic slot. At 3,000 m and nominal sound speed 1,500 m/s, a round trip alone takes about four seconds; three sequential targets need at least twelve seconds plus processing/guard time. A one-second aggregate polling promise would therefore be physically impossible for that schedule, despite advertised update rates under other geometry. Distinct codes, controlled slots, acoustic guard bands and collision-safe vehicle spacing protect the observations that the model assumes.

For search resource e, target k and action a, form expected gain g_{eka} from that target's surviving distribution and completion cost τ_{eka}. Binary assignment variables satisfy one task per resource, compatible depth/tooling, one simultaneous survey per interference cell and collision/time-window constraints. Maximizing $\sum g_{eka} x_{eka}$ for fixed fleet effort is a small assignment problem; a fractional gain/effort objective requires enumerating or solving its own ratio, not replacing it silently with total gain.

A computed two-resource/two-target check uses three shared cells A/B/C. Target masses are $(0.60, 0.25, 0.15)$ and $(0.10, 0.35, 0.55)$; conditional detection rows are $(0.8, 0.7, 0.5)$ and $(0.7, 0.8, 0.9)$; effort rows are $(1.0, 1.5, 2.0)$ and $(2.0, 1.2, 1.0)$ h. Exhaustive enumeration of the six nonconflicting assignments selects A for the first and C for the second: 0.975 expected discoveries for 2.0 fleet-hours, or 0.4875 per fleet-hour. The next best assignment, A/B, yields

0.760 for 2.2 fleet-hours. The numerical example is conditional and small, but it verifies that identity-specific probability and shared-cell exclusion actually enter resource choice. A full fleet schedule must also include the travel from each resource's current position.

8 Recommended operating policy and limitations

MCMS should preserve independent acoustic contact, transmit depth/fault/life-support status, and carry a rapid calibrated host tracking package. For deep operation, secure a tested search/intervention contract with compatible ship, launch equipment, operators and observation latency. Use a joint fault/current/detectability ensemble, deploy at the highest feasible near-term gain, and update on each actual miss rather than marking a cell "cleared." Negative searches should trigger new environmental information and another sensing modality as residual mass moves toward difficult regimes.

The conditional calculation gives faster discovery than both the compact lawnmower and the stronger same-domain cumulative-coverage route. Its narrower 11.1-point final-discovery and 1.27 h restricted-mean advantages support retaining survival-aware search in this scenario; they do not prove global superiority. The forcing challenge identifies the main limitation: inaccurate currents and incorrect contact/detection mechanisms can overwhelm a sophisticated route. Fault frequencies, sonar hazard and real standby performance remain uncalibrated. Life support, certified lifting/hatch compatibility, hull integrity and sea-state constraints are prerequisites for an approval envelope. Search capability should never be used to authorize a depth at which no credible timely recovery exists.

Memorandum to the Government of Greece

To: Ministry responsible for maritime safety, Government of Greece
From: Team 7391861
Subject: Conditions for approving MCMS tourist submersible operations
Date: 11 October 2026

We recommend conditional approval within a demonstrated operating envelope, with depth limited by recovery capability. Finding a lost vehicle cannot protect passengers unless a compatible system can recover them before life support expires. Deep Ionian operations require this evidence alongside depth-rated search equipment.

MCMS should carry an independently powered emergency acoustic beacon on every submersible and a calibrated tracking/telemetry receiver on the host. Every dive should log synchronized position estimates and uncertainty, depth and depth change, ballast/leak status, thruster commands and life-support margins. Timed escalation and acoustic reacquisition must use the age of the last trusted fix, including missed intervals and processing delay.

Our model distinguishes seabed contact, suspension and continued propulsion after communication failure. These states require different search depths and routes. Dispatch should use their location distribution, revised after each actual unsuccessful survey. A miss does not clear a cell when terrain or poor acoustics can cause repeated failures. A route responding only to misses can be prepared in advance; new current measurements or fault reports justify fresh dispatch decisions.

In a fully specified synthetic deep-Ionian scenario, this approach produced a 70.8% conditional chance of finding the vehicle within six hours and 81.2% within eighteen hours. A fixed systematic search with the same speeds, descent delay, survey effort, detector and return reserve produced 26.1% and 50.2%. A stronger cumulative-coverage plan using the same reachable field, clock and return reserve found 70.2% by eighteen hours, with restricted mean time 7.7 h versus 6.5 h. The remaining advantage is 11.1 percentage points; this is a comparison of declared policies, not a separate measurement of online feedback or a causal split of the old gain. These are comparative model results, not certified real-incident rescue rates. Stronger currents in another scenario reduced the eighteen-hour result to 47.3%, making local current measurements and acoustic trials essential.

The host should own rapid acoustic tracking. Deep imaging and intervention may be contracted with specified availability, response times, maintenance, operators, transport, compatible launch arrangements and prompt search observations. The economic assessment should include capital, servicing, training, standby, deployment and vessel usage. MCMS should provide current quotations and response commitments; verified commercial prices were unavailable to this study.

The approval evidence should include fault demonstrations, missed-contact drills, sonar detection trials in representative seabed and midwater conditions, and rehearsal of the complete recovery chain. A quotation or a manufacturer's maximum depth alone is insufficient evidence of readiness.

Memorandum continued: approval conditions and emergency response

A published deep survey AUV can operate to 6,000 m, and a scientific intervention ROV example reaches 6,500 m. Neither specification proves that it can bring passengers to safety. The published NATO docking rescue vehicle is limited to 610 m and needs a compatible escape hatch and substantial support. It cannot be assumed to rescue occupants in our 3,000 m scenario or to be available under a commercial Greek arrangement.

MCMS should demonstrate an engineered emergency ascent or recovery method at each proposed depth. This must address hull integrity, ballast release, seabed entrapment, certified lifting points and rigging, lifting loads, controlled ascent and surface handling. If a passenger transfer method is proposed, its pressure rating, hatch compatibility, medical and decompression arrangements must be established. A search result should be accepted as a recovery task only after the system identifies the vehicle and confirms the relevant condition.

The emergency clock should begin at the last trusted status, not when an outside vessel arrives. The response plan must show that detection, intervention, extraction and a reserve fit within measured life-support endurance. A technically superior rescue tool that mobilizes too late offers no protection in this time budget. Our calculation explicitly integrates discovery time with a hypothetical compatible deep recovery clock, while leaving its unproved intervention success separate. The regulator should require a credible response margin, including adverse weather, maintenance outages and more than one incident.

During an incident, the host should activate emergency acoustics, preserve the latest telemetry, declare the missing-contact event according to the agreed threshold, and call the contracted response team promptly. Search equipment should be sent to the feasible location/depth with the highest expected gain per elapsed time. After each survey, the supervisor should record coverage, detection quality and candidates, update the location distribution, and obtain further current or depth information if results contradict the initial picture. Repeated misses should prompt an independent sensor or a revised fault model, rather than increased confidence in the same unsuccessful pattern.

For multiple tourist vehicles, unique identifiers and an acoustic schedule are essential. Search and intervention resources must be jointly allocated; anonymous contacts, interference and conflicting routes can otherwise undermine the individual search plans. A new destination, including the Caribbean, needs its own current and density information, bathymetry, launch/weather constraints, port access and recovery agreement. The equations transfer; the Greek scenario's numbers do not.

We propose an approval envelope tied to verified depth, environmental limits, crew readiness, tracking quality and recovery time. MCMS should submit the trial data, current quotations, maintenance records, contracts and recovery demonstration for that envelope. Where an acceptable recovery chain has not been demonstrated, restrict depth or suspend the corresponding operation. This approach gives the company a concrete path to approval while keeping the decision grounded in passenger recovery rather than a search probability alone.

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Report on Use of AI

System and date. OpenAI Codex, GPT-6, 11 October 2026. This is an autonomous AI-authored development solution to the original 2024 MCM B problem. The AI formulated the models, selected and read the general sources, wrote all code, executed computations, interpreted the conditional results, and wrote and typeset the technical report and memorandum. No human solving or human validation is claimed.

Instructions and output. The execution assignment requested a complete scientific solution to Locate, Prepare, Search and Extrapolate, an exactly two-page government memo, a first complete draft followed by substantive improvement, bounded reproducible computation, negative-search updating, equal physical search comparisons, and truthful source and AI records. The original problem and its four pages were read and visually inspected. Project prior contact with the problem is known; no earlier same-problem solution or reference paper was supplied or read during this episode. Only frozen methodology/build support and general official ocean/equipment/search sources were used.

Available record. The complete output is this report with its local code, parameter file, result and verification JSON, source ledger, run snapshots and build receipts. The local live progress record documents decisions, actual failures and version boundaries. These files are execution evidence; they do not constitute an automatically captured verbatim chat transcript. A complete exact prompt/output transcript is not available in the delivered artifact, so no reconstructed transcript is presented as complete.

Verification and limitations. Programmatic checks include exact Bayes arithmetic, an analytic covariance special case, physical boundaries and independently implemented route/hazard replay. The same AI author reviewed the mathematical assumptions, synchronized the prose and viewed the rendered PDF; this is author verification, not an independent human assessment. Official source claims were read at the cited locations. Exact procurement prices, real incident currents, calibrated failure priors, detector performance and real recovery compatibility remain unavailable. The stated discovery rates are conditional synthetic results and do not authorize a rescue operation or prove regulatory compliance.

Scientific development and feedback. A coordinating AI agent authorized one fixed-budget coverage comparison. The current host label is `gpt-6.1-sol`, `ultra` effort; the service build is unverified. The AI retrieved three project lessons, applied persistent-quality and equal-service conditions, and rejected an unrelated lesson. It implemented one coverage route, separately integrated the frozen routes, synchronized the manuscript and inspected its PDF. A fresh same-family AI reviewer then assessed the complete scientific successor. The author corrected missed-report last-fix age and recovery-clock unit conversion, checked exact rational boundaries, and inspected the rebuilt PDF. Frozen search results and five scientific figures were preserved. No human solving or validation is claimed. Actual new messages and tool records are captured from the explicit child-thread JSONL; inherited coordinator context is marked separately. Snapshots may end during an active turn and do not certify complete historical transcripts. Exact token/credits and service-side costs remain unavailable.